

# Shift11

## AI-Powered Football Momentum Attribution

### Phase-Wise Development Plan

**Core principle:** deterministic analytics establish what happened; AI explains why it mattered.

**Differentiator:** detect the shift → attribute the player → reconstruct the evidence → explain why it mattered.

### Phase 1 — Project Foundation

- Python project structure, virtual environment, Git/GitHub workflow.
- Teams, players, positions, formations, match metadata and event schema.
- Seeded synthetic 90-minute match generator with reproducible JSON output.
- Event validation before analytics.

**Exit criterion:** Reproducible synthetic match can be generated, saved, loaded and validated.

### Phase 2 — Deterministic Momentum Analytics

- Match-state model and event aggregation.
- 0–100 score: territory 20%, possession/progression 25%, pressure/recoveries 20%, chance creation 25%, possession control 10%.
- 5-minute rolling window, updated every 30 seconds, with recency weighting.
- Shift rule: momentum +15, opponent advantage +10, persistence  $\geq 60$  seconds, and  $\geq 3$  meaningful supporting events.
- Hysteresis/cooldown to prevent duplicate/noisy markers.

**Exit criterion:** Controlled synthetic sequences produce expected momentum and shift objects.

### Phase 3 — Momentum Attribution & Evidence Chains

- Player contribution scoring and primary/secondary attribution.
- Roles: Momentum Catalyst, Progression Driver, Chance Creator, Defensive Trigger, Finisher.
- Coherent 5–7 event evidence chain per shift.
- Structured, verifiable fact pack for UI/AI.

**Exit criterion:** Known catalyst ranks highest and its evidence chain explains the shift.

### Phase 4 — Tactical Evidence & Validation

- Before/after metrics and replayable event references.
- Validate event taxonomy and roles against pressing, progression, recoveries, chance creation and finishing.
- Consistency and auditability checks.

**Exit criterion:** Every displayed shift traces back to raw events and measurable changes.

### Phase 5 — Grounded Microsoft AI Layer

- Pass only verified fact packs to the AI layer.
- Microsoft Agent Framework for agent/orchestration; Microsoft Foundry for model/runtime path.
- Prompt constraints prohibit invented events, statistics or unsupported claims.
- Evidence-linked tactical explanations.

**Exit criterion:** AI remains grounded when tested against known fact packs.

## Phase 6 — Intelligence & Personalization

- Casual, Tactical and Player-focused explanation modes.
- Questions such as what changed, who drove it, why it mattered, and what to watch next.
- All responses remain anchored to verified facts.

**Exit criterion:** Audience modes change presentation, not underlying facts.

## Phase 7 — Interactive Demo UI

- Momentum timeline with shift markers.
- Drill-down: timestamp, momentum change, confidence, before/after metrics, contributors, roles and evidence.
- Interactive pitch/player highlighting where practical.
- Simple judging flow: Shift → Attribution → Evidence → Explanation.

**Exit criterion:** A judge can understand a shift and its cause from one drill-down.

## Phase 8 — Azure Deployment & Microsoft Integration

- Package application for Azure.
- If time/access permits, host Agent Framework agent through Microsoft Foundry Hosted Agents.
- Secure environment variables and Azure identity/authentication.
- Keep local execution as a judging fallback.

**Exit criterion:** Microsoft AI path works end-to-end without making cloud deployment a single point of failure.

## Phase 9 — Testing, Evaluation & Submission

- Unit tests for momentum, shift detection, attribution, evidence ordering and fact packs.
- Grounding tests for AI explanations.
- Controlled synthetic matches with known catalysts.
- README, architecture, demo script/video and submission materials.

**Exit criterion:** Reproducible end-to-end demo with documented tests.

## Minimum Judging Scope

The core demo must include: seeded synthetic events; deterministic momentum timeline; reliable shift detection; primary/secondary player attribution; contribution scores and roles; 5–7 event evidence chain; before/after metrics; grounded Microsoft Agent Framework + Foundry explanation; interactive shift drill-down; and tests.

Advanced player-strength/weakness analysis, richer pitch visuals and additional Azure services are enhancements and must not block the core demo.